

CODSOFT Artificial Intelligence Internship

Chatbot – Learn Mate AI

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> **INTERNSHIP DOMAIN:** - [Artificial Intelligence](#)

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INTRODUCTION

LearnMate AI is a Python-based educational learning assistant developed to help students learn technical topics, practice quizzes, prepare for viva questions, and explore career opportunities. The application provides information on subjects such as Python, Artificial Intelligence, Machine Learning, Data Science, Cyber Security, and Cloud Computing.

The main objective of this project is to make learning simple, interactive, and accessible. It also motivates students through inspirational quotes and helps them improve their technical knowledge and career awareness.

OBJECTIVE

- To develop an educational learning assistant using Python.
- To provide basic knowledge of technical topics such as Python, AI, ML, Data Science, Cyber Security, and Cloud Computing.
- To help students assess their knowledge through quizzes.
- To support viva and interview preparation.
- To provide career guidance for different technology fields.
- To encourage continuous learning through motivational quotes.

TECHNOLOGY USED

TECHNOLOGIES USED

- Programming Language: Python 3
- Development Environment: Visual Studio Code (VS Code)
- Concepts Used:
 - Functions
 - Conditional Statements (if-elif-else)
 - Loops
 - Lists
 - User Input Handling
- Interface: Command Line Interface (CLI)
- Operating System: Windows 10/11

FEATURES

- Topic Learning Module
 - Provides basic information on Python, AI, ML, Data Science, Cyber Security, and Cloud Computing.
- Quiz Module
 - Allows users to answer multiple-choice questions and receive instant feedback.
- Viva Practice Module
 - Helps students prepare for viva examinations with common questions and answers.
- Career Guidance Module
 - Provides career roadmaps for AI Engineer, Data Scientist, Web Developer, Cyber Security Analyst, and Cloud Engineer.
- Daily Motivation Module
 - Displays motivational quotes to encourage learning and self-improvement.

- User-Friendly Menu
- Simple menu-driven interface for easy navigation.

SYSTEM REQUIREMENTS

Hardware Requirements:

- Processor: Intel Core i3 or above
- RAM: 4 GB or above
- Storage: 100 MB Free Space

Software Requirements:

- Operating System: Windows 10/11
- Python 3.x
- Visual Studio Code (VS Code)
- Command Prompt / Terminal

ALGORITHM

Step 1: Start the program.

Step 2: Display the main menu options.

Step 3: Accept the user's choice.

Step 4: If the user selects "Learn a Topic", display information about the selected topic.

Step 5: If the user selects "Take a Quiz", ask quiz questions and display answers.

Step 6: If the user selects "Viva Practice", show viva questions and answers.

Step 7: If the user selects "Career Guidance", display the selected career roadmap. Step

8: If the user selects "Daily Motivation", display motivational quotes.

Step 9: Repeat the process until the user selects the Exit option. Step

10: Display a thank you message and terminate the program.

SOURCE CODE

```
def show_menu():
    print("\n" + "=" * 45)
    print("      Welcome to LearnMate AI")
    print("=" * 45)
    print("1. Learn a Topic")
    print("2. Take a Quiz")
    print("3. Viva Practice")
    print("4. Career Guidance")
    print("5. Daily Motivation")
    print("6. Exit")

def learn_topic():
    topic = input("\nWhich topic would you like to learn? (Python/AI/ML/ds/cyber/cloud): ").lower()

    if topic == "python":
        print("\nPython is one of the easiest programming languages.") print("It
        is widely used in web development, AI, and data science.")

    elif topic == "ai":
        print("\nAI stands for Artificial Intelligence.")
        print("It enables computers to think and solve problems like humans.")

    elif topic == "ml":
        print("\nMachine Learning is a part of AI.")
        print("It helps systems learn from data without explicit programming.")

    elif topic == "ds":
        print("\nData science is the study of data ") print("It
        helps in finding useful insights from data.")

    elif topic == "cyber":
        print("\nCyber Security protects systems and data.")
        print("It helps prevent hacking and cyber attacks.")

    elif topic == "cloud":
        print("\nCloud Computing provides services over the internet.")
```

```
        print("Examples are Google Drive and AWS.")

    else:
        print("\nSorry, this topic is not available right now.")

def quiz():
    print("\n--- Quick Quiz ---")

    print("\nWhich company created Python?")
    print("A. Microsoft")
    print("B. Google")
    print("C. Python Software Foundation")
    answer = input("Your Answer: ").upper()

    if answer == "C":
        print("Correct!")
    else:
        print("Wrong! Correct Answer: Python Software Foundation")

    print("\nWhat does AI stand for?")
    print("A. Artificial Intelligence")
    print("B. Automatic Internet")
    print("C. Artificial Interface")
    answer = input("Your Answer: ").upper()

    if answer == "A":
        print("Correct!")
    else:
        print("Wrong! Correct Answer: Artificial Intelligence")

    if answer == "A":
        print("Great! That's the correct answer.")

    print("\nWhat does ML stand for?")
    print("A. Machine Learning")
    print("B. Main Logic")
    print("C. Machine Language")
    answer = input("Your Answer: ").upper()

    if answer == "A":
        print("Correct!")
    else:
        print("Wrong! Correct Answer: Machine Learning")
```

```
print("\nWhat is Data Science used for?")
print("A. Data Analysis")
print("B. Gaming")
print("C. Drawing")
answer = input("Your Answer: ").upper()
```

```
if answer == "A":
    print("Correct!")
else:
    print("Wrong! Correct Answer: Data Analysis")
```

```
print("\nWhich is an example of Cloud Computing?") print("A.
Google Drive")
print("B. Paint")
print("C. Calculator")
answer = input("Your Answer: ").upper()
```

```
if answer == "A":
    print("Correct!")
else:
    print("Wrong! Correct Answer: Google Drive")
```

```
print("\nWhat is the purpose of Cyber Security?") print("A.
Protecting Data")
print("B. Playing Games")
print("C. Watching Videos")
answer = input("Your Answer: ").upper()
```

```
if answer == "A":
    print("Correct!")
else:
    print("Wrong! Correct Answer: Protecting Data")
```

```
def viva():
    print("\n--- Viva Practice ---")
    print("Question: What is Python?")
    input("\nPress Enter to see the answer...")
    print("Answer: Python is a high-level and easy-to-learn programming language.")

    print("\nQ7. What is the full form of CPU?")
    input("Press Enter to see the answer...")
    print("Answer: CPU stands for Central Processing Unit.")
```

```
print("\nQ8. What is HTML?") input("Press  
Enter to see the answer...")  
print("Answer: HTML is used to create the structure of web pages.")
```

```
print("\nQ9. What is CSS?") input("Press  
Enter to see the answer...")  
print("Answer: CSS is used to style web pages.")
```

```
print("\nQ10. What is a Database?")  
input("Press Enter to see the answer...")  
print("Answer: A database is an organized collection of data.")
```

```
def career_roadmap(): print("\nChoose  
a Career Path:") print("1. AI  
engineer")  
print("2. Data scientist")  
print("3. Web developer")  
print("4. Cyber security analiyst")  
print("5. cloud engineer")  
  
choice = input("Enter your choice: ") if  
  
choice == "1":  
    print("\nAI ENGINEER ROADMAP")  
    print("Step 1: Learn Python") print("Step  
2: Learn Machine Learning") print("Step  
3: Learn Deep Learning") print("Step 4:  
Build AI Projects") print("Step 5: Apply  
for Internships")  
  
elif choice == "2":  
    print("\nDATA SCIENTIST ROADMAP")  
    print("Step 1: Learn Python")  
    print("Step 2: Learn Statistics")  
    print("Step 3: Learn Data Analysis")  
    print("Step 4: Work on Projects")  
    print("Step 5: Build Portfolio")  
  
elif choice == "3":  
    print("\nWEB DEVELOPER ROADMAP")  
    print("Step 1: Learn HTML")  
    print("Step 2: Learn CSS")  
    print("Step 3: Learn JavaScript")
```

```
print("Step 4: Build Websites")
print("Step 5: Create Portfolio")
```

```
elif choice == "4":
    print("\nCYBER SECURITY ROADMAP")
    print("Step 1: Learn Networking")
    print("Step 2: Learn Linux") print("Step
3: Learn Ethical Hacking") print("Step 4:
Practice Security Tools") print("Step 5:
Get Certifications")
```

```
elif choice == "5":
    print("\nCLOUD ENGINEER ROADMAP")
    print("Step 1: Learn Linux")
    print("Step 2: Learn Networking")
    print("Step 3: Learn AWS or Azure")
    print("Step 4: Build Cloud Projects")
    print("Step 5: Apply for Jobs")
```

```
else:
    print("Invalid choice.")
```

```
def motivation():
    quotes = [
        "Success comes from consistent effort.",
        "Keep learning, keep growing.",
        "Every expert was once a beginner."
    ]
```

```
print("\nMotivation of the Day:")
for quote in quotes:
    print("•", quote)
```

```
# Main Program
```

```
while True:
```

```
    show_menu()
```

```
    user_choice = input("\nSelect an option (1-6): ") if
```

```
    user_choice == "1":
        learn_topic()
```

```
elif user_choice == "2": quiz()

elif user_choice == "3": viva()

elif user_choice == "4":
    career_roadmap()

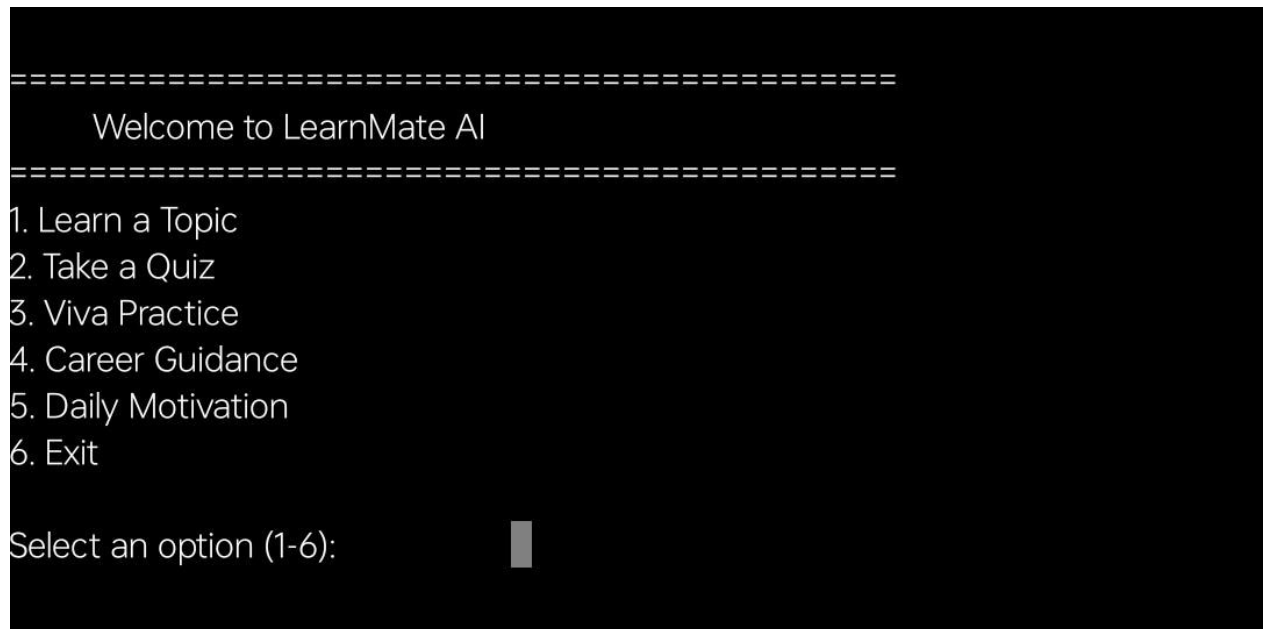
elif user_choice == "5":
    motivation()

elif user_choice == "6":
    print("\nThank you for using LearnMate AI.")
    print("Have a great learning journey!") break

else:
    print("\nPlease enter a valid option.")
```

OUTPUT

Figure: - 1 Main Menu Of LearnMate AI

A screenshot of a terminal window with a black background and white text. The text displays a welcome message for 'LearnMate AI' enclosed in dashed lines. Below this, a numbered list from 1 to 6 offers options: 'Learn a Topic', 'Take a Quiz', 'Viva Practice', 'Career Guidance', 'Daily Motivation', and 'Exit'. At the bottom, a prompt 'Select an option (1-6):' is followed by a small grey rectangular cursor.

```
=====
Welcome to LearnMate AI
=====
1. Learn a Topic
2. Take a Quiz
3. Viva Practice
4. Career Guidance
5. Daily Motivation
6. Exit

Select an option (1-6): █
```

Figure: - 2 Quick Quiz Module Output

--- Quick Quiz ---

Which company created Python?

- A. Microsoft
- B. Google
- C. Python Software Foundation

Your Answer: c

Correct!

What does AI stand for?

- A. Artificial Intelligence
- B. Automatic Internet
- C. Artificial Interface

Your Answer: a

Correct!

Great! That's the correct answer.

What does ML stand for?

- A. Machine Learning
- B. Main Logic
- C. Machine Language

Your Answer: a

Correct!

Figure: - 3 Career Guidance Module Output

Choose a Career Path:

1. AI engineer
2. Data scientist
3. Web developer
4. Cyber security analyst
5. cloud engineer

Enter your choice: 1

AI ENGINEER ROADMAP

Step 1: Learn Python

Step 2: Learn Machine Learning

Step 3: Learn Deep Learning

Step 4: Build AI Projects

Step 5: Apply for Internships

•The above screenshot shows the main interface of LearnMate AI. The application successfully displays all available options, including topic learning, quiz practice, viva preparation, career guidance, and daily motivation. Users can select any option according to their learning requirements.

FUTURE ENHANCEMENTS

- Add a Graphical User Interface (GUI) for a better user experience.
- Include more learning topics and quiz questions.
- Add a score tracking system to monitor student performance.
- Implement voice-based interaction.
- Provide personalized learning recommendations.
- Integrate AI features for smarter responses and guidance.

CONCLUSION

LearnMate AI is a simple and effective educational learning assistant developed using Python. It helps students learn technical topics, practice quizzes, prepare for viva examinations, explore career paths, and stay motivated. The project successfully demonstrates the use of Python programming concepts to create an interactive and user-friendly learning application.

REFERENCES

1. Python Official Documentation – <https://docs.python.org>
2. Visual Studio Code Documentation
3. Online Learning Resources and Tutorials
4. CodSoft Artificial Intelligence Internship Guidelines
5. Educational Content on Python, AI, Machine Learning, Data Science, Cyber Security, and Cloud Computing.

RESULT

The LearnMate AI application was successfully developed and tested using Python. The system correctly provides learning content, conducts quizzes, supports viva preparation, offers career guidance, and displays motivational quotes. All modules function as expected and provide an interactive learning experience for students. The project successfully achieved its objectives and demonstrated the practical implementation of Python programming concepts.